

Low noise amplifier parts guide

Revision 4

- An LNA can be constructed according to the circuit diagram using small, surface-mount circuit elements and a pre-fabricated printed circuit board.
- The pre-fabricated printed circuit board designed for this LNA can be purchased from Oshpark at https://oshpark.com/shared_projects/zvqWjxWi
- Most of the circuit parts can be ordered from Digikey at [digikey.com](https://www.digikey.com), as listed in the table below. The RF shield can be ordered from Leader Tech at [leadertechinc.com](https://www.leadertechinc.com).
- The amplifier chips, U1 – U5, can be obtained from Mini-Circuits at [minicircuits.com](https://www.minicircuits.com). You can request free “samples” from the web page.

Parts are grouped by type. Original component values and part numbers are retained.

Circuit parts

Circuit Diagram Part	Description	Digikey Part #	Quantity
C1, C5, C6	27 pF	712-1334-2-ND	3
C2, C3, C7, C8, C9, C10, C11, C12, C13, C14	330 pF	478-1181-2-ND	10
C4	4700 pF	399-14206-2-ND	1
C15	6.8 pF	712-1364-2-ND	1
C16	10 μ F	399-8153-1-ND	1
R1	33 k Ω	311-33.0KHRTR-ND	1
R2	27 Ω	P26.1HTR-ND	1
R3, R8	50 Ω	RR08Q49.9DTR-ND	2
R4, R13	43 Ω	311-43.0HRTR-ND	2
R5	4.7 k Ω	541-4.70KHTR-ND	1
R6	17.4 Ω	541-17.4HTR-ND	1
R7, R14	5.8 Ω	311-5.76HRTR-ND	2
R9, R10	287 Ω	541-287HTR-ND	2
R11, R12, R15, R16	870 Ω	311-866HRTR-ND	4
L1, L3, L4, L5, L6	100 nH	490-5664-2-ND	5

Circuit Diagram Part	Description	Digikey Part #	Quantity
L2	10 nH	490-5648-2-ND	1
IC1	LM2940 voltage regulator	LM2937IMPX-5.0/NOPBTR-ND	1
X1	CONN SMA RCPT STR 50 OHM PCB	WM5543-ND	1
X2	CONN SMA JACK R/A 50 OHM PCB	931-1361-ND	1

Other manufacturers

Circuit Diagram Part	Description	Manufacturer	Part No.
U1, U3	Amplifier	Mini-Circuits	GALI 39+
U2	Amplifier	Mini-Circuits	SAV-541+
U4, U5	LTCC Band Pass Filter	Mini-Circuits	BFCN-1445+
Circuit Board Shielding	RF shield for circuit board	Leader Tech	14R-CBSA-0.5X2.75X0.13